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Measuring the Quality of Network Configurations Generated using LLM Models

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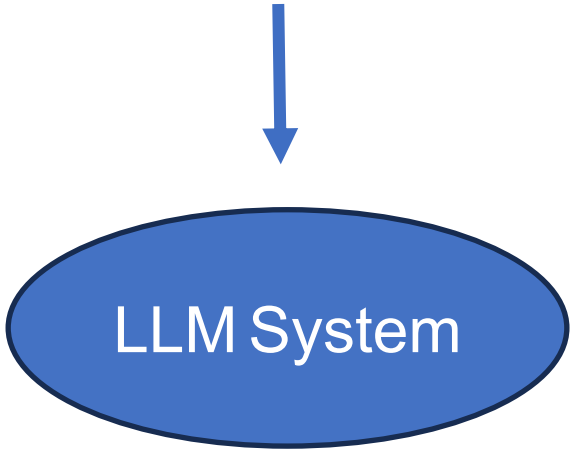
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Network Configuration Generated by LLM Models

Configuration task

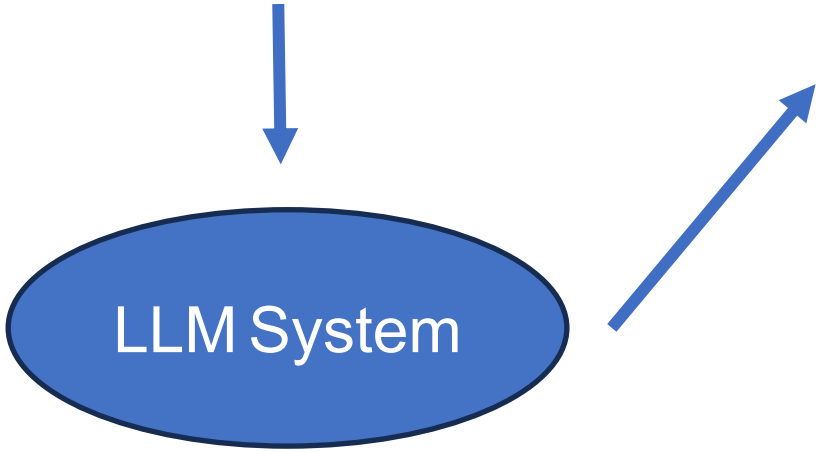
“Generate the configuration for a Cisco ISR 1200 with an IP address of 192.168.1.1. Create a static route to the destination 10.10.10.0/24 via the ge0 interface.”



Network Configuration Generated by LLM Models

Configuration task

“Generate the configuration for a Cisco ISR 1200 with an IP address of 192.168.1.1. Create a static route to the destination 10.10.10.0/24 via the ge0 interface.”




```
Router(config)# hostname isr1200

isr1200(config)# interface FastEthernet0
isr1200(config-if)# ip address 192.168.1.1 255.255.255.0

isr1200(config-if)# exit

isr1200(config)# ip default-gateway 192.168.1.254

isr1200(config)# ip classless

isr1200(config)# ip route 10.10.10.0 255.255.255.0 ge0

isr1200(config)# line con 0
isr1200(config-line)# login local

isr1200(config-line)# exit

isr1200(config)# enable password ciscolab

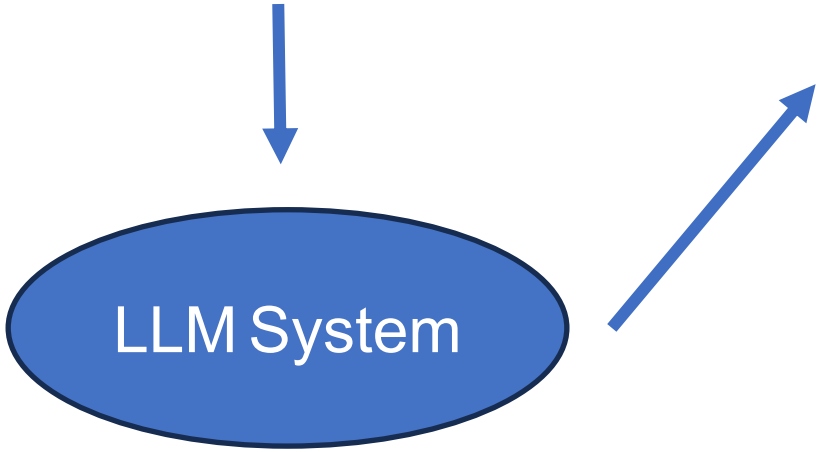
isr1200(config)# service password-recovery

isr1200(config)# end
```

Network Configuration Generated by LLM Models

Configuration task

"Generate the configuration for a Cisco ISR 1200 with an IP address of 192.168.1.1. Create a static route to the destination 10.10.10.0/24 via the ge0 interface."




```
Router(config)# hostname isr1200

isr1200(config)# interface FastEthernet0
isr1200(config-if)# ip address 192.168.1.1 255.255.255.0
                    no shutdown
isr1200(config-if)# exit

isr1200(config)# ip default-gateway 192.168.1.254

isr1200(config)# ip classless

isr1200(config)# ip route 10.10.10.0 255.255.255.0 ge0

isr1200(config)# line con 0
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isr1200(config-line)# exit

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isr1200(config)# end
```

Mask not specified

Missed command

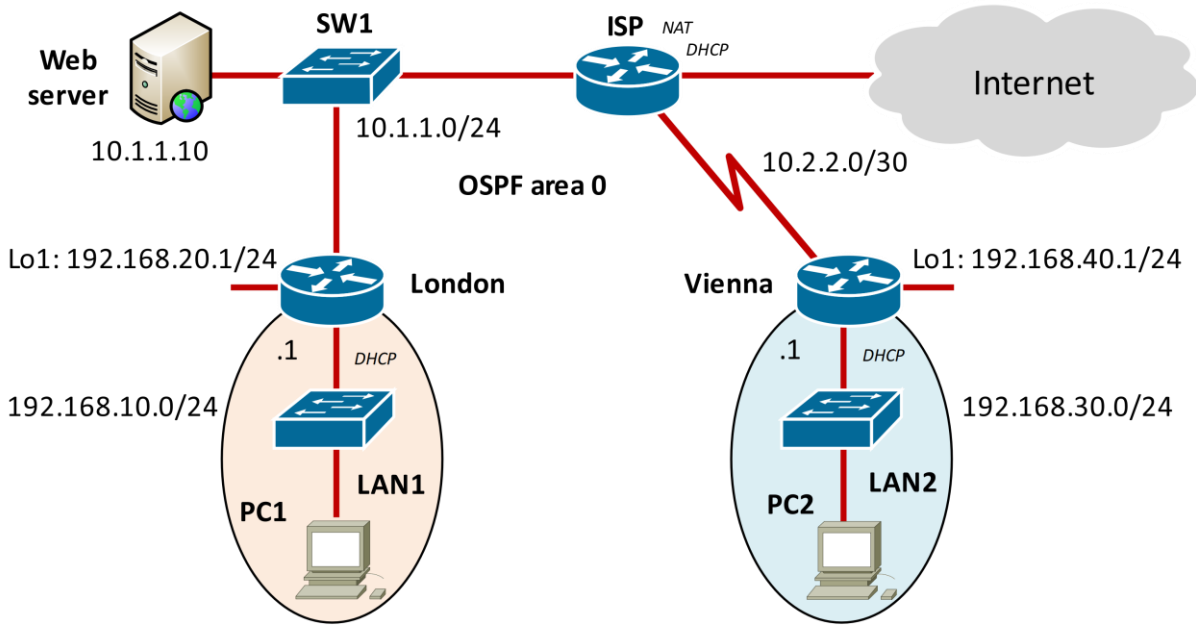
Not needed

Not requested

How to Measure the Quality of Generated Network Configurations

Statistical metrics such as BLEU, ROUGE and Levenshtein distance are insufficient.

- Even small changes to the config file can render the entire configuration incorrect.
- There can be multiple ways to configure a requested task.
- Tasks that are not specified precisely may produce correct but unwanted output.
- Default parameters can be listed explicitly in the configuration file.
- The correct placement of commands is important.



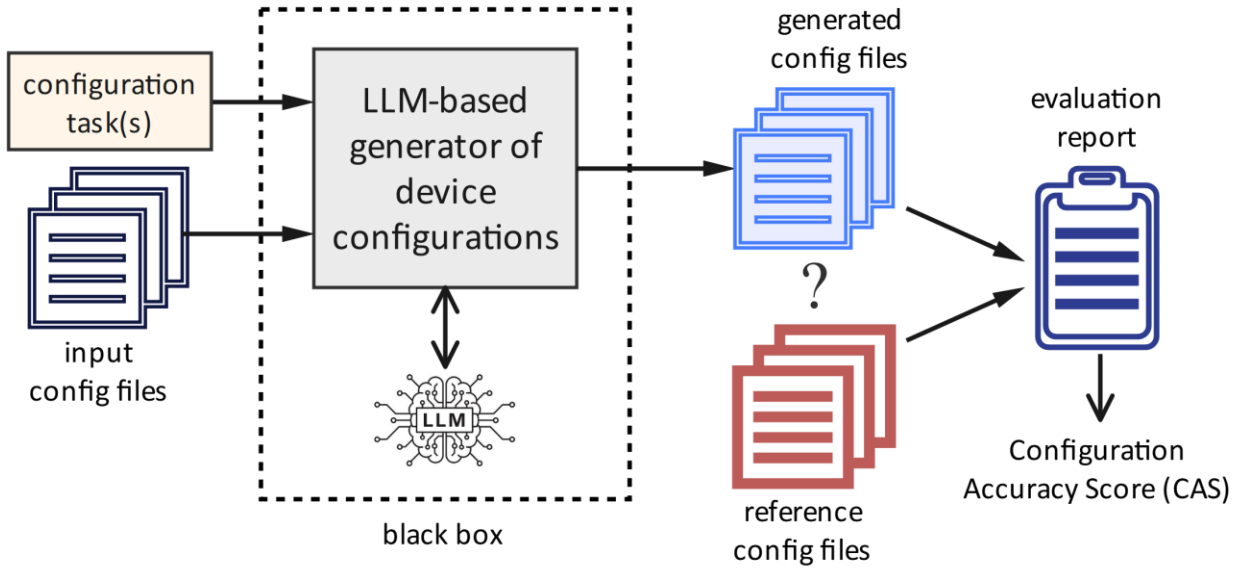
Proposed Multi-dimensional Scoring System

1) Static comparison with the reference configuration

CAS – Configuration Accuracy Score

- The generated configuration is compared with the reference configuration.
- CAS represents the syntactic similarity of the configurations.
- The CAS score for task T = *normalised number of additions (a), missed commands (m), and alternatives (t)*

$$CAS = 1 - \frac{w_a a + w_m m + w_t t}{N}$$



- N = number of lines in the reference configuration
- The weights reflect the relative importance of the changes.
- A CAS score of 1 means the configurations are equivalent.

- Changes can be harmful (CAS = 0) or harmless (CAS > 0).

Proposed Multi-dimensional Scoring System

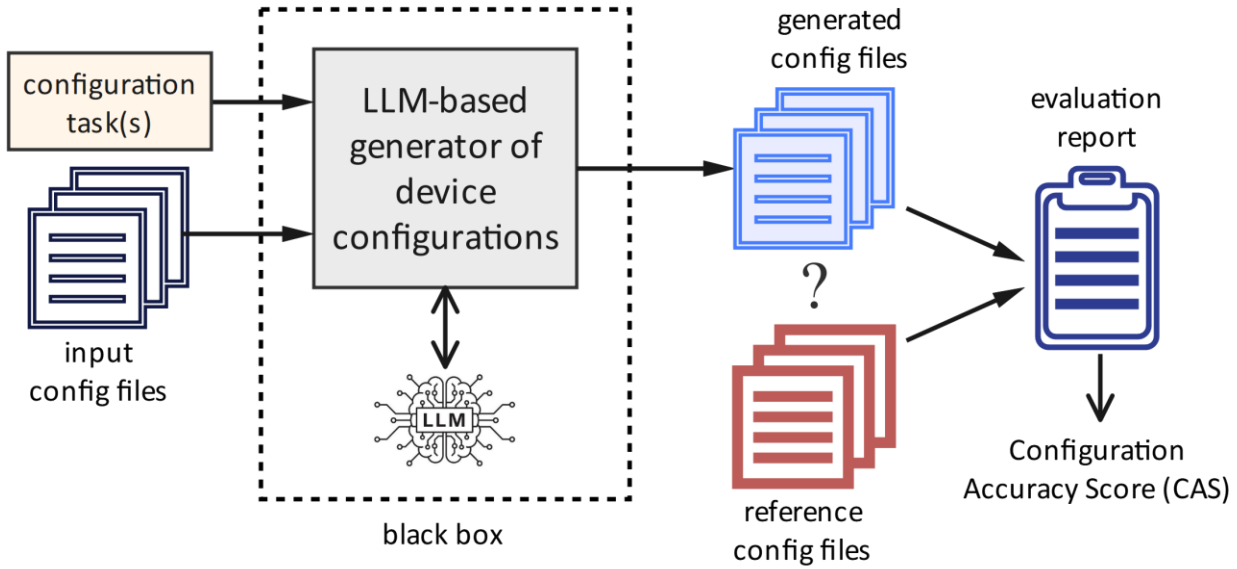
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Limitations

- The reference configuration is not always available.
- It is not easy to compare alternative configurations.

Proposed Multi-dimensional Scoring System

2) Semantic and regression validation

WFC – Weighted Feature Coverage

- WFC quantifies the coverage of the feature tests $t \in F$, indicating the correctness of the generated configuration c^{gen}.

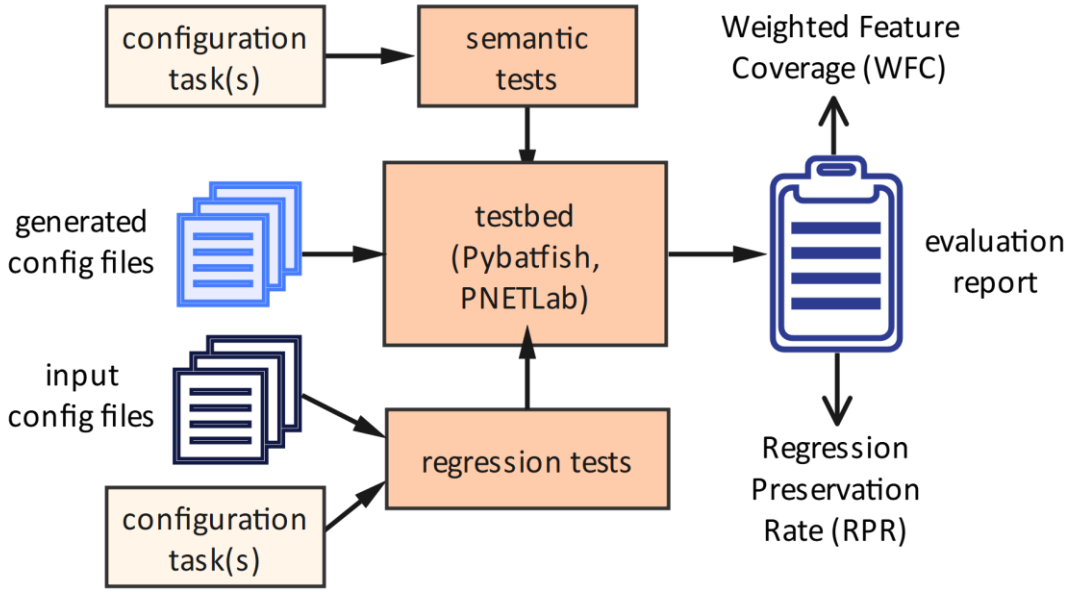
$$WFC = \frac{\sum_{t \in F} w(t)\, passed(t, c^{gen})}{\sum_{t \in F} w(t)}$$

- WFC = 1 means that all the feature tests were successful.

RPR – Regression Preservation Rate

- RPR is the proportion of regression tests $t \in R$ that still pass on c^{gen}.

$$RPR = \frac{|\{t \in R \mid passed(t, c^{gen}) = 1\}|}{|R|}$$



Proposed Multi-dimensional Scoring System

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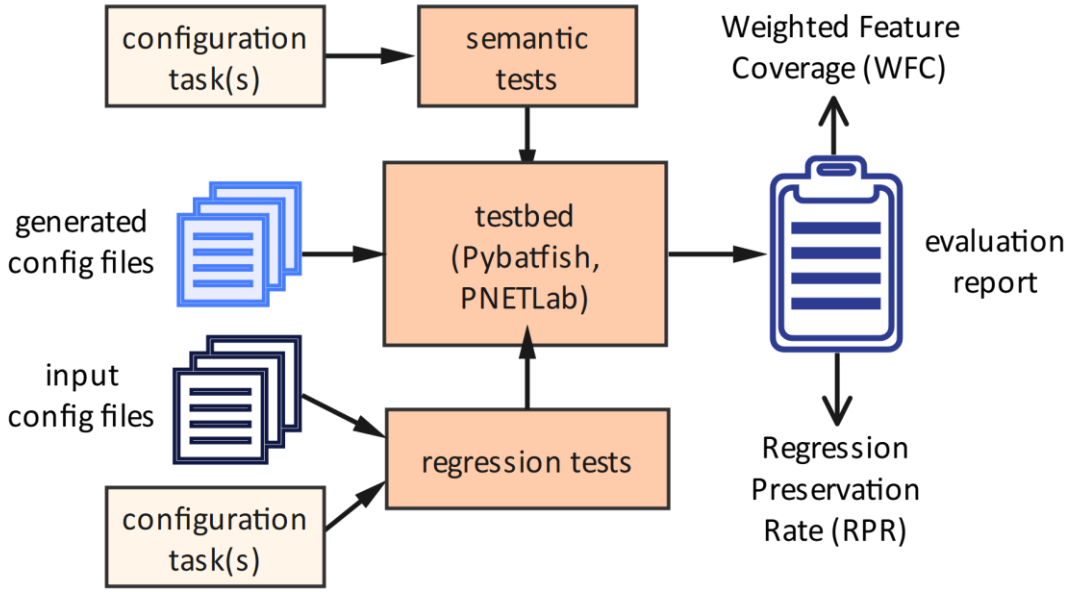
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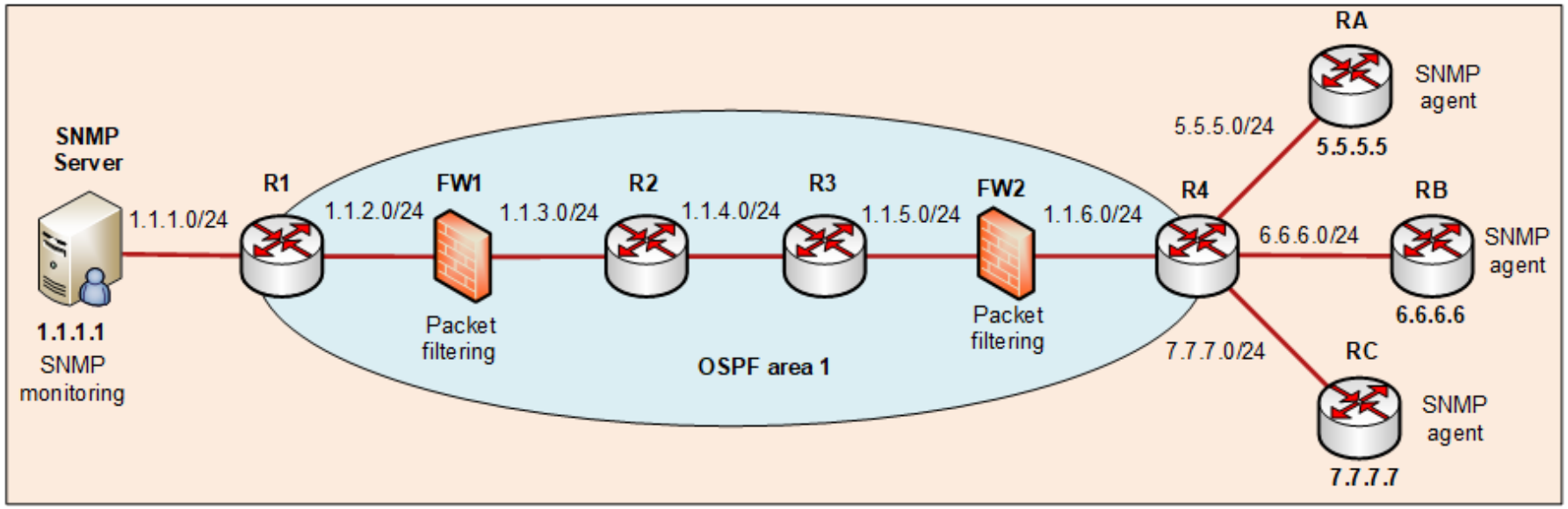


Limitations

- Semantic tests cannot cover all aspects of the generated configuration.
- Passing tests do not guarantee a flawless configuration.

Experiments: SNMP Scenario

Topology:



Task description:

- Configure the SNMP agents on routers RA, RB and RC.
- Set the community string to “att-monitoring”.
- Restrict access to the SNMP service to the 1.1.1.1 host only.
- Enable traps to be sent to the monitoring server.
- Configure the firewalls to enable SNMP monitoring.
- Update the ACLs on routers FW1 and FW2 to allow SNMP communication.

Experiments: SNMP Scenario

Evaluation of generated configuration using multi-dimensional scoring system

Model	CAS (*)	WFC	RPR
GPT-5	79.6%	100%	100%
DeepSeek	74.5%	58.3%	100%
Grok	89.4%	91.7%	100%
GPT-OSS (run 1)	52.0%	58.3%	100%
GPT-OSS (run 2)	68.5%	41.7%	100%
GPT-OSS (run 3)	61.4%	100%	100%
GPT-OSS (run 4)	64.7%	50%	100%
GPT-OSS (run 5)	83.2%	66.7%	97.8%

(*) CAS computed without classifying changes as harmless or harmful

- *CAS = similarity with the reference configuration*
- *WFC = success rate of feature tests*
- *RPR = preservation of previously established behaviour*

- The proposed quantitative metrics cover both the static and dynamic aspects of the generated configuration.
- If the evaluation fails, this indicates possible problems with the generated configuration.
- The metrics allow comparing LLM models on the same configuration tasks.

Automated Evaluation of Functional and Regression Tests

Pybatfish Verifier

- Analyses device configurations offline, without running devices; multi-vendor.
- **WFC – 12 feature tests:** 4 checks × 3 SNMP-enabled routers (RA, RB, RC).
- **RPR – 9 regression tests:** evaluated on the original and generated configuration of each modified device.
- Reachability queries verify end-to-end behaviour across FW1 and FW2.

Available at https://pybatfish.readthedocs.io/

```
# Feature tests (shown for RA)
snmp = viModel().nodes['ra'].snmpServer
assert 'att-monitoring' in snmp.communities   # community string
assert '1.1.1.1' in snmp.hosts                # traps

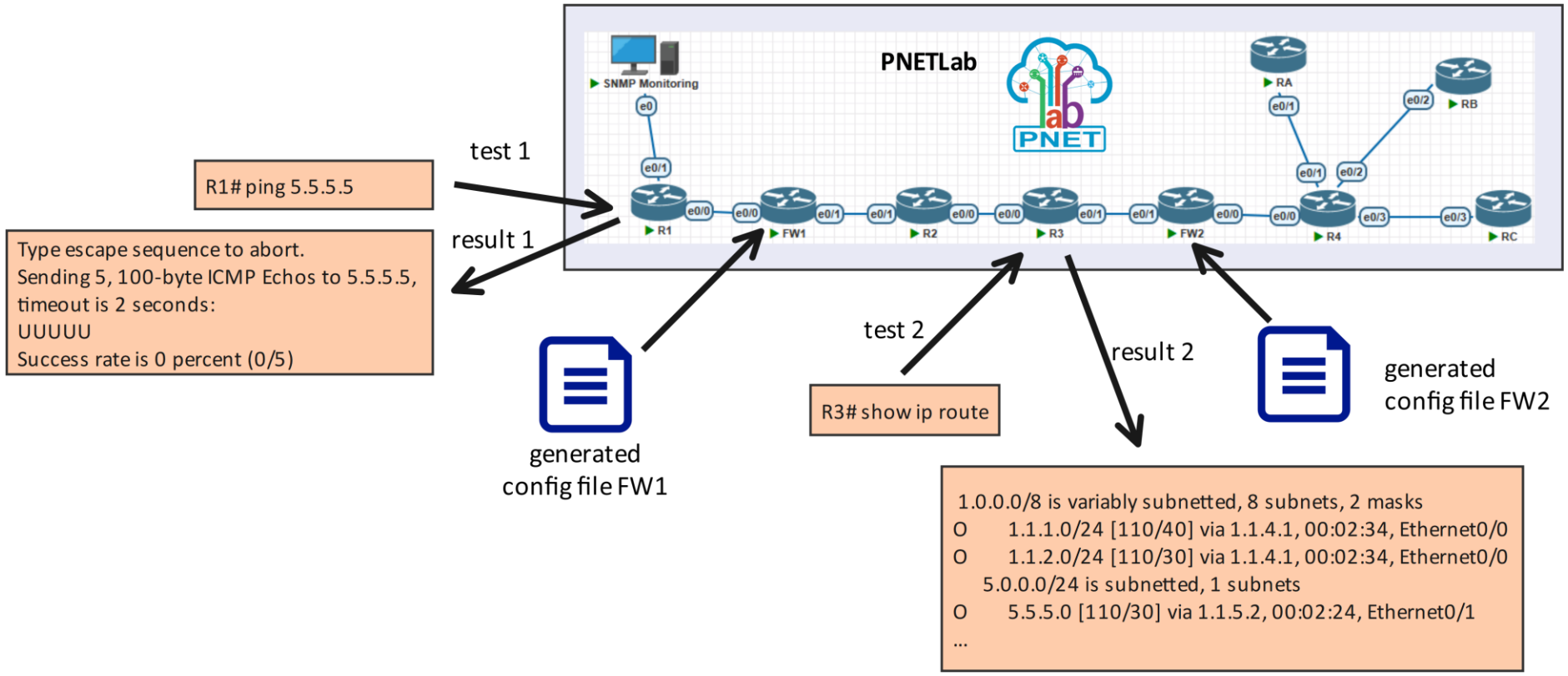
# SNMP allowed from the monitoring server ...
assert reachability(srcIps='1.1.1.1', dstIps='5.5.5.5', applications='SNMP', actions='Failure').empty

# ... and blocked from any other host
assert not reachability(srcIps='1.1.1.100', dstIps='5.5.5.5',
                        applications='SNMP', actions='Failure').empty
```

Automated Evaluation of Functional and Regression Tests

Packet Network Emulator Tool Lab (PNETLab)

- A platform that emulates network topology and communication using virtualised network devices.



- A generated configuration with testing commands can be sent to PNETLab via the API.
- The output of the testing commands is then evaluated against the expected test results.

Available at https://pnetlab.com

Evaluation Scenarios: Topology, Tasks, Reference Configurations

Scenario	Number of devices	Evaluated tasks for tested scenarios
ACL	3 routers	IP addressing, DHCP server, SSH access, OSPF routing, ACL filtering
GRE	3 routers, 1 switch	IP addressing, DHCP server, NAT translation, GRE tunnel, OSPF routing
HSRP	5 routers, 2 switches	IP addressing, static routing, NAT translation, DHCP, HSRP configuration
IPv6	3 routers, 1 switch	IPv4 and IPv6 addressing, DHCPv4 and DHCPv6, IPv6 tunnelling, OSPF and RIPng routing
BGP	3 routers	IP addressing, BGP routing, route redistribution
IPsec	2 routers	IP addressing, static routing, ISAKMP configuration, key distribution, ACL, IPsec tunnelling
Monitoring	3 routers, 2 switches	NTP synchronisation, NAT, DHCP, Syslog, SNMPv2 and SNMPv3, CDP
MPLS	6 routers	IP addressing, OSPF routing, MPLS encapsulation, VRF, BGP routing
SNMP	7 routers, 2 firewalls	SNMP, ACLs

Available at https://github.com/MichalRozsival/configuration-agent